Task 1: Web Scraping and Custom Dataset Creation

Data Analysis

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Course:
Data Analysis

Task:
Task 1 – Web Scraping and Custom Dataset Creation

Tools Used:
Python, Requests, BeautifulSoup, Pandas, Matplotlib

Dataset Source:
Books to Scrape

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1. Introduction

Web scraping is a data collection technique used to automatically extract structured information from web pages. In this task, Python was used to collect publicly available book information from the Books to Scrape website. The extracted information was processed, cleaned, transformed, and stored as a structured dataset.

The project demonstrates a complete basic data analysis workflow, starting from web data collection and ending with dataset creation, analysis, and visualization. A total of 100 book records were collected from the first five pages of the website.

2. Objectives

The main objectives of this task were:

- To understand the basic concepts of web scraping.
- To extract relevant information from public web pages using Python.
- To understand the HTML structure of a web page.
- To use Python libraries such as Requests and BeautifulSoup for data extraction.
- To organize the extracted information into a structured dataset.
- To clean and transform the collected data into an analysis-ready format.
- To create a customized dataset by adding a derived Price Category feature.
- To perform basic statistical analysis on the collected data.
- To visualize the analyzed data using a bar chart.

3. Tools and Technologies

- The following tools and technologies were used to complete the task:

Tool/Technology	Purpose

Python	Main programming language used for web scraping and data processing
Requests	Used to send HTTP requests and retrieve web pages
BeautifulSoup	Used to parse HTML and extract required information
Pandas	Used to create, clean, transform, and analyze the dataset
Matplotlib	Used to create the data visualization
Visual Studio Code	Used as the development environment
CSV	Used to store the final structured dataset

- The project was implemented using Python in Visual Studio Code.

4. Website Selected

- For this task, the public practice website **Books to Scrape** was selected as the data source. The website contains book-related information such as book titles, prices, availability status, and ratings.
- The website was selected because it provides publicly accessible data in a structured HTML format, making it suitable for practicing web scraping techniques without requiring authentication or access to private information.
- The first five catalogue pages were used for data collection.

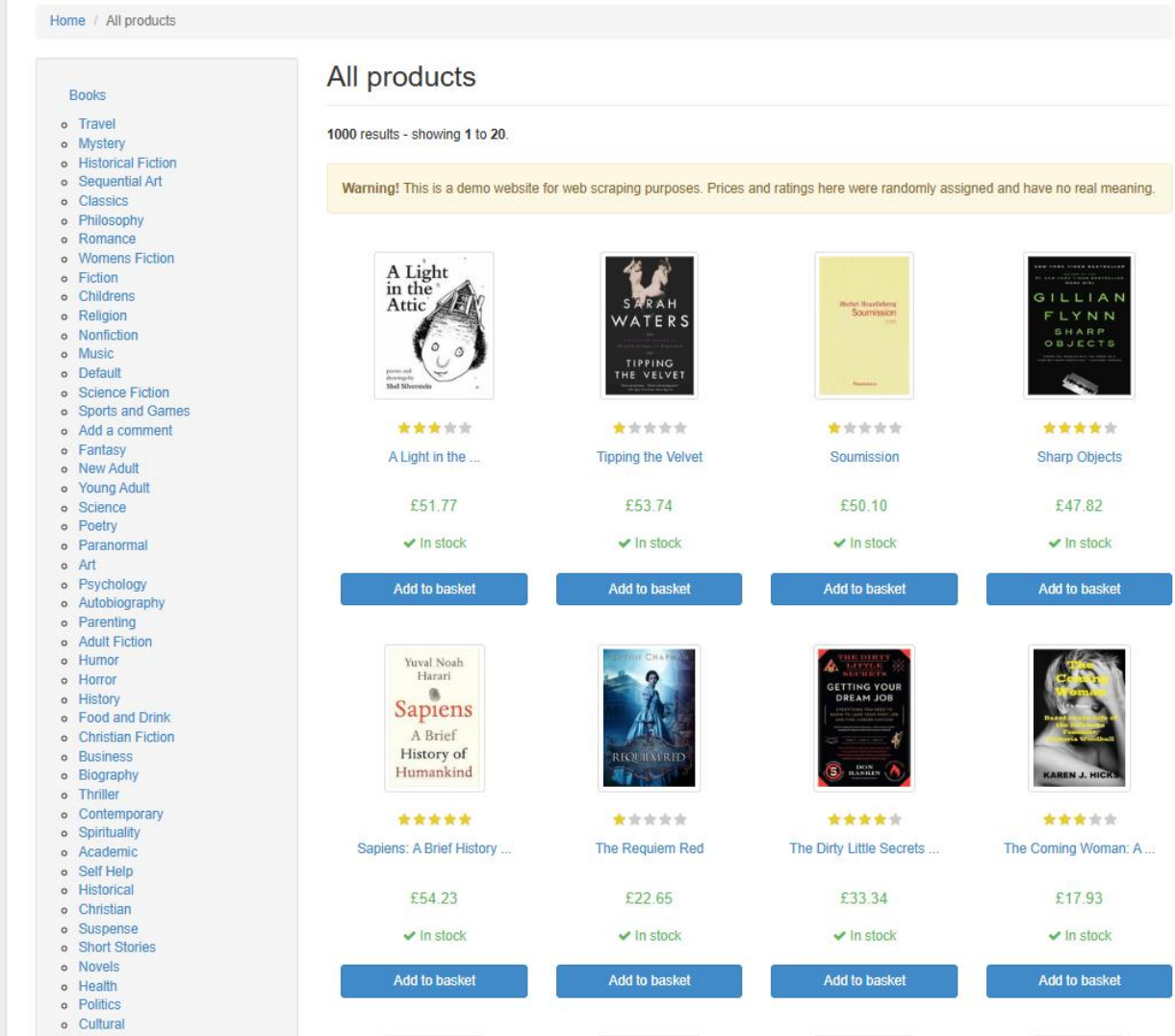


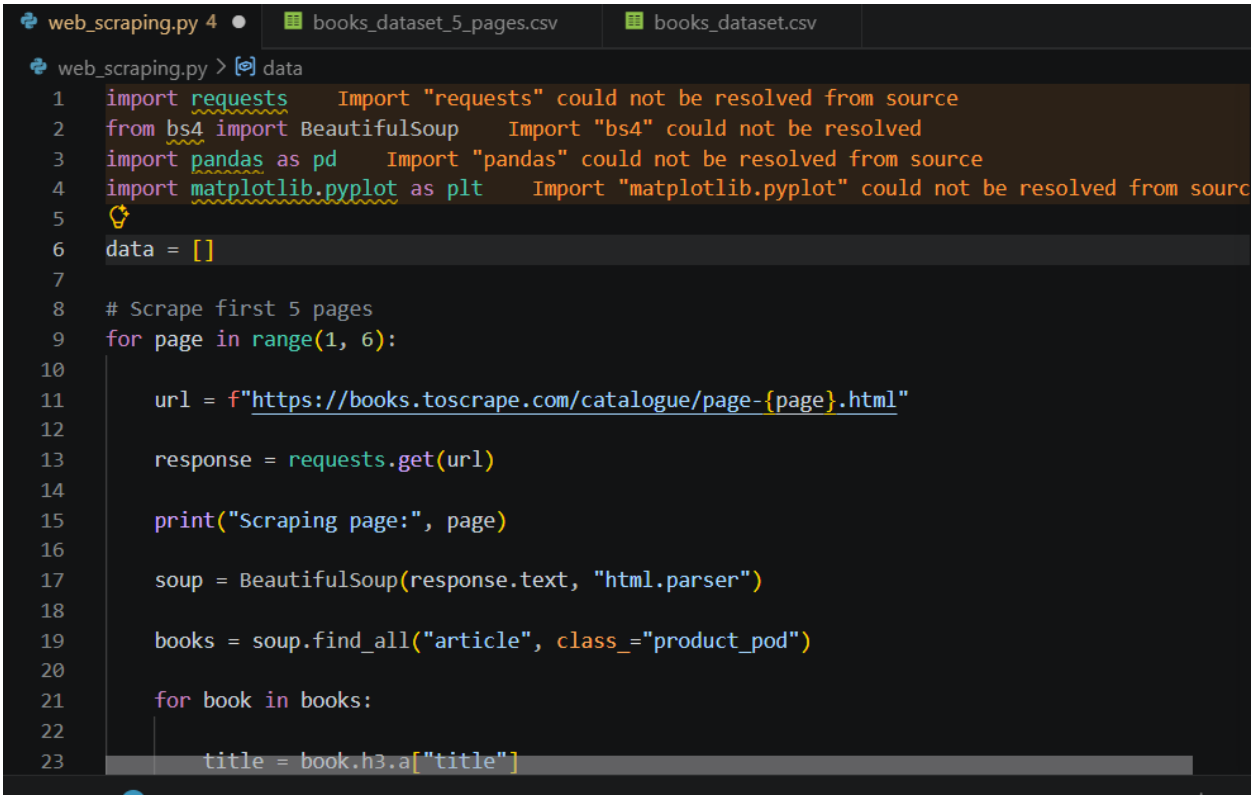
Figure 1: Books to Scrape website used as the data source

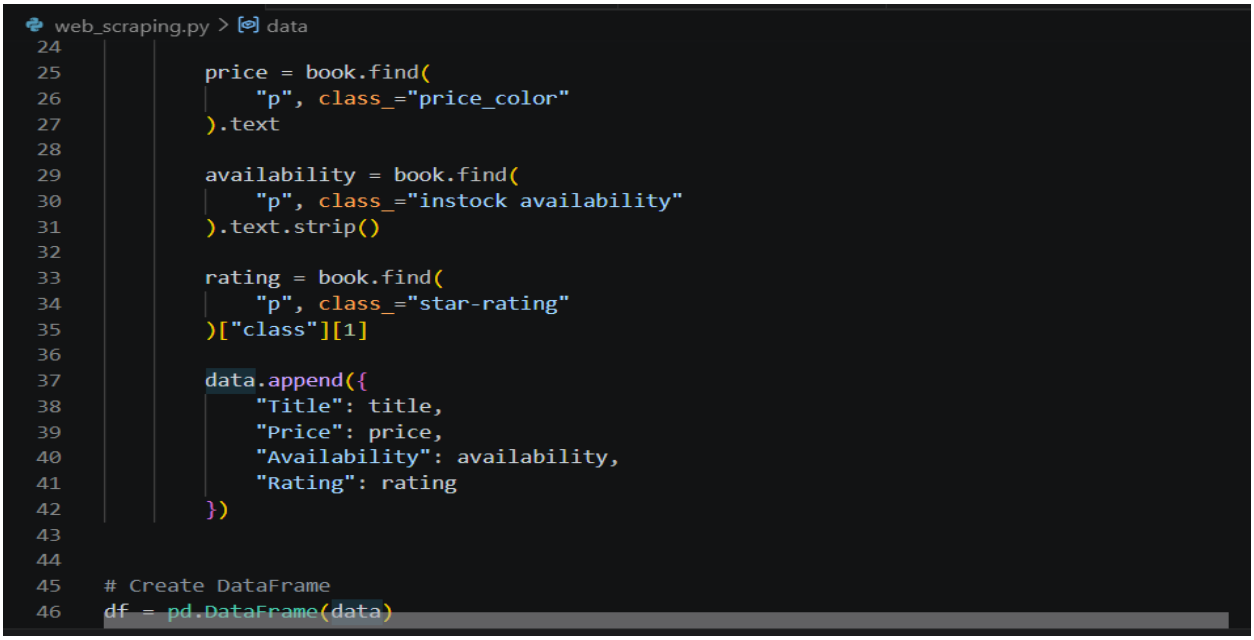
5. Web Scraping Methodology

The web scraping process was implemented using Python. The Requests library was used to send HTTP requests to the website and retrieve the HTML content of each catalogue page. BeautifulSoup was then used to parse the HTML structure and locate the required book information.

The scraper was programmed to navigate through the first five catalogue pages. For each page, the program identified the book elements and extracted the title, price, availability, and rating.

The extracted information was temporarily stored in a Python list and was later converted into a Pandas DataFrame for further processing and analysis.


web_scraping.py 4
books_dataset_5_pages.csv
books_dataset.csv
web_scraping.py > data
import requests Import "requests" could not be resolved from source
from bs4 import BeautifulSoup Import "bs4" could not be resolved
import pandas as pd Import "pandas" could not be resolved from source
import matplotlib.pyplot as plt Import "matplotlib.pyplot" could not be resolved from sourc
data = []
Scrape first 5 pages
for page in range(1, 6):
url = f"https://books.toscrape.com/catalogue/page-{page}.html"
response = requests.get(url)
print("Scraping page:", page)
soup = BeautifulSoup(response.text, "html.parser")
books = soup.find_all("article", class_="product_pod")
for book in books:
title = book.h3.a["title"]


web_scraping.py > data
price = book.find(
"p", class_="price_color"
).text
availability = book.find(
"p", class_="instock availability"
).text.strip()
rating = book.find(
"p", class_="star-rating"
)["class"][1]
data.append({
"Title": title,
"Price": price,
"Availability": availability,
"Rating": rating
})
Create DataFrame
df = pd.DataFrame(data)

```python
# Clean Price column
df["Price"] = (
    df["Price"]
    .str.extract(r"(\d+\.\d+)")[0]
    .astype(float)
)


# Create custom Price Category
def price_category(price):

    if price < 20:
        return "Low"

    elif price < 40:
        return "Medium"

    else:
        return "High"

```

```python
    df["Price Category"] = df["Price"].apply(price_category)


    # Save dataset
    df.to_csv("books_dataset_5_pages.csv", index=False)


    # Display basic analysis
    print("\nScraping completed!")
    print("Total records:", len(df))

    print("\nFirst 5 records:")
    print(df.head())

    print("\nDataset Shape:")
    print(df.shape)

    print("\nAverage Price:")
    print(df["Price"].mean())

    print("\nMaximum Price:")
    print(df["Price"].max())

    print("\nMinimum Price:")
    print(df["Price"].min())

    print("\nRating Count:")
    print(df["Rating"].value_counts())

    print("\nPrice Category Count:")
    print(df["Price Category"].value_counts())


    # Create visualization
    category_counts = df["Price Category"].value_counts()

    category_counts.plot(kind="bar")

    plt.title("Books by Price Category")
    plt.xlabel("Price Category")
    plt.ylabel("Number of Books")

    plt.tight_layout()
```

```
113
114    plt.savefig("price_category_chart.png")
115
116    plt.show()
```

Figure 2: Python web scraping implementation in Visual Studio Code

6. Data Extraction

The scraper extracted four main attributes from each book:

- **Title** – The name of the book.
- **Price** – The listed price of the book.
- **Availability** – The availability status of the book.
- **Rating** – The star rating assigned to the book.

The scraper processed five catalogue pages and collected a total of 100 book records. The extracted data was stored in a Pandas DataFrame before further cleaning and transformation.

```
Scraping page: 1
Scraping page: 2
Scraping page: 3
Scraping page: 4
Scraping page: 5

Scraping completed!
Total records: 100
```

```
Dataset Shape:
(100, 5)

Average Price:
34.560700000000004

Maximum Price:
58.11

Minimum Price:
10.16
```

Figure 3: Successful extraction of 100 book records from five web pages

7. Data Cleaning and Transformation

After extracting the data, the Price column required cleaning because the original web data contained currency symbols and text formatting. The numeric portion of each price was extracted and converted into a floating-point numeric data type.

This transformation made the Price column suitable for mathematical calculations and statistical analysis.

The cleaned dataset was then checked to ensure that the extracted records were properly structured and ready for further analysis.

8. Custom Dataset Creation

To make the dataset more useful for analysis, a new derived feature called **Price Category** was created. The books were categorized according to their prices using the following rules:

- **Low:** Price below 20
- **Medium:** Price from 20 to below 40
- **High:** Price equal to or above 40

This additional feature transformed the original scraped dataset into a customized dataset suitable for categorical analysis and visualization.

The final dataset contains 100 records and five columns: Title, Price, Availability, Rating, and Price Category.

	A	B	C	D	E
1	Title	Price	Availabilit	Rating	Price Cate
2	A Light in the Attic	51.77	In stock	Three	High
3	Tipping the Velvet	53.74	In stock	One	High
4	Soumission	50.1	In stock	One	High
5	Sharp Objects	47.82	In stock	Four	High
6	Sapiens: A Brief History of Humankind	54.23	In stock	Five	High
7	The Requiem Red	22.65	In stock	One	Medium
8	The Dirty Little Secrets of Getting Your Dream Job	33.34	In stock	Four	Medium
9	The Coming Woman: A Novel Based on the Life of the Infamous Fe	17.93	In stock	Three	Low
10	The Boys in the Boat: Nine Americans and Their Epic Quest for Gol	22.6	In stock	Four	Medium
11	The Black Maria	52.15	In stock	One	High
12	Starving Hearts (Triangular Trade Trilogy, #1)	13.99	In stock	Two	Low
13	Shakespeare's Sonnets	20.66	In stock	Four	Medium
14	Set Me Free	17.46	In stock	Five	Low
15	Scott Pilgrim's Precious Little Life (Scott Pilgrim #1)	52.29	In stock	Five	High
16	Rip it Up and Start Again	35.02	In stock	Five	Medium
17	Our Band Could Be Your Life: Scenes from the American Indie Und	57.25	In stock	Three	High
18	Olio	23.88	In stock	One	Medium
19	Mesaerion: The Best Science Fiction Stories 1800-1849	37.59	In stock	One	Medium
20	Libertarianism for Beginners	51.33	In stock	Two	High
21	It's Only the Himalayas	45.17	In stock	Two	High
22	In Her Wake	12.84	In stock	One	Low
23	How Music Works	37.32	In stock	Two	Medium
24	Foolproof Preserving: A Guide to Small Batch Jams, Jellies, Pickles,	30.52	In stock	Three	Medium
25	Chase Me (Paris Nights #2)	25.27	In stock	Five	Medium

	A	B	C	D	E
78	Saga, Volume 6 (Saga (Collected Editions) #6)	25.02	In stock	Three	Medium
79	Saga, Volume 5 (Saga (Collected Editions) #5)	51.04	In stock	Two	High
80	Reskilling America: Learning to Labor in the Twenty-First Century	19.83	In stock	Two	Low
81	Rat Queens, Vol. 3: Demons (Rat Queens (Collected Editions) #11-	50.4	In stock	Three	High
82	Princess Jellyfish 2-in-1 Omnibus, Vol. 01 (Princess Jellyfish 2-in-1	13.61	In stock	Five	Low
83	Princess Between Worlds (Wide-Awake Princess #5)	13.34	In stock	Five	Low
84	Pop Gun War, Volume 1: Gift	18.97	In stock	One	Low
85	Political Suicide: Missteps, Peccadilloes, Bad Calls, Backroom Hijin	36.28	In stock	Two	Medium
86	Patience	10.16	In stock	Three	Low
87	Outcast, Vol. 1: A Darkness Surrounds Him (Outcast #1)	15.44	In stock	Four	Low
88	orange: The Complete Collection 1 (orange: The Complete Collecti	48.41	In stock	One	High
89	Online Marketing for Busy Authors: A Step-By-Step Guide	46.35	In stock	One	High
90	On a Midnight Clear	14.07	In stock	Three	Low
91	Obsidian (Lux #1)	14.86	In stock	Two	Low
92	My Paris Kitchen: Recipes and Stories	33.37	In stock	Two	Medium
93	Masks and Shadows	56.4	In stock	Two	High
94	Mama Tried: Traditional Italian Cooking for the Screwed, Crude, V	14.02	In stock	Four	Low
95	Lumberjanes, Vol. 2: Friendship to the Max (Lumberjanes #5-8)	46.91	In stock	Two	High
96	Lumberjanes, Vol. 1: Beware the Kitten Holy (Lumberjanes #1-4)	45.61	In stock	Three	High
97	Lumberjanes Vol. 3: A Terrible Plan (Lumberjanes #9-12)	19.92	In stock	Two	Low
98	Layered: Baking, Building, and Styling Spectacular Cakes	40.11	In stock	One	High
99	Judo: Seven Steps to Black Belt (an Introductory Guide for Beginne	53.9	In stock	Two	High
100	Join	35.67	In stock	Five	Medium
101	In the Country We Love: My Family Divided	22	In stock	Four	Medium
102					

Figure 5: Final customized dataset containing the Price Category feature

9. Data Analysis

Basic descriptive analysis was performed on the collected dataset using Pandas. The analysis included the first five records, dataset dimensions, average price, maximum price, minimum price, and the frequency distribution of book ratings.

The dataset contains 100 records and five columns. The Price column was analyzed to calculate its average, maximum, and minimum values. In addition, the frequency of each rating category was calculated to understand the distribution of book ratings.

The Price Category feature was also analyzed to determine the number of books classified as Low, Medium, and High priced.

```
First 5 records:
                                Title  Price Availability Rating Price Category
0                A Light in the Attic  51.77     In stock  Three           High
1                  Tipping the Velvet  53.74     In stock    One           High
2                          Soumission  50.10     In stock    One           High
3                       Sharp Objects  47.82     In stock   Four           High
4  Sapiens: A Brief History of Humankind  54.23     In stock   Five           High
```

```
Price Category Count:
Price Category
High      40
Medium    34
Low       26
Name: count, dtype: int64
```

```
Rating Count:
Rating
Three    22
One      22
Five     19
Two      19
Four     18
Name: count, dtype: int64
```

Figure 6: Descriptive analysis and category counts generated using Pandas

10. Data Visualization

To visually represent the distribution of books across the different price categories, a bar chart was created using Matplotlib.

The visualization represents the number of books classified as Low, Medium, and High priced. This provides a simple visual comparison of the different price categories within the collected dataset.

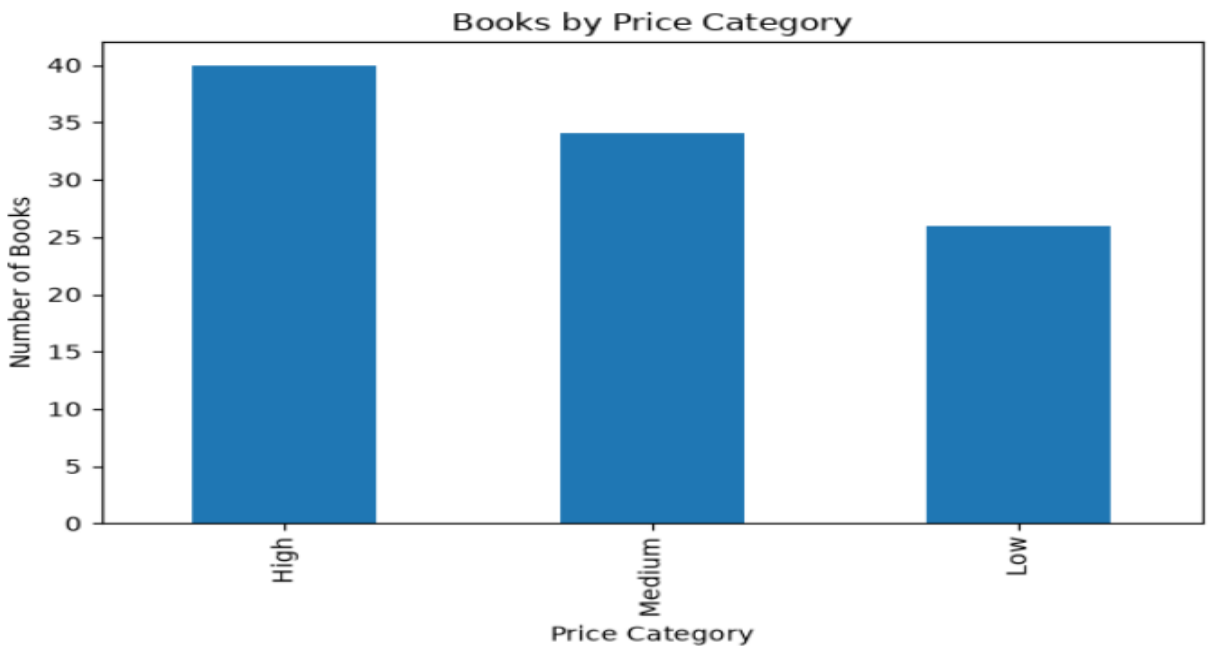


Figure 7: Distribution of books by price category

11. Results

The web scraping process successfully collected 100 book records from the first five pages of the Books to Scrape website.

The final dataset contains five attributes: Title, Price, Availability, Rating, and Price Category. The Price values were successfully cleaned and converted into numeric values, allowing statistical calculations to be performed.

The dataset was further analyzed using descriptive statistics and rating frequency counts. A custom Price Category feature was also created to classify books into Low, Medium, and High price groups.

Finally, a bar chart was generated to visualize the distribution of books across the three price categories. The results demonstrate that web scraping can be used to collect, transform, analyze, and visualize publicly available web data effectively.

12. Conclusion

This task provided practical experience in collecting and processing web-based data using Python. The Requests and BeautifulSoup libraries were used to extract structured information from the Books to Scrape website, while Pandas was used for data cleaning, transformation, and analysis.

A total of 100 records were collected from five web pages and transformed into a customized dataset by introducing the Price Category feature. Basic statistical analysis and visualization were then performed using Pandas and Matplotlib.

Overall, the task demonstrated the complete workflow of web scraping, data cleaning, custom dataset creation, analysis, and visualization. It also provided practical understanding of how publicly available web data can be converted into a structured dataset for analytical purposes.